Phase 4 (part 1): First-Order Autonomous ODEs

1.] Let $y(t)$ be the quantity or concentration of a substance at time t. Then $y(t)$ experiences exponential growth or decay if it satisfies the following initial value problem:

$$\frac{dy}{dt} = ky \qquad y(0) = y_0,$$

where $(0, y_0)$ is a known point on the graph of $y(t)$ and k is the per capita growth/decay rate. Here, $y(0)$ is called the initial data. Solve this IVP.

$$\int \frac{1}{y}\,dy = \int k\,dt$$

$$\ln|y| = kt + C$$

$$|y| = e^{kt+C}$$

$$y = \pm e^{kt+C}$$

$$y = \underbrace{\pm e^{C}}\, e^{kt}$$

$$\boxed{y(t) = Ce^{kt}}$$

gen sol

$$y(0) = Ce^{k\cdot 0}$$

$$y_0 = C$$

$$\boxed{y(t) = y_0 e^{kt}}$$

unique sol

2.] According to Newton's Law of Cooling, the rate of change of temperature of a cooling object is proportional to the temperature *difference* between the object and the surrounding medium. That is,

$$\frac{dy}{dt} = -k\,(y - y_s)\,, \qquad y(0) = y_0$$

where $y(t)$ is the temperature of the object at time t, and y_s is the temperature of the surroundings, assumed to be constant. Here, $k > 0$. Solve this IVP. What is the equilibrium temperature?

$$\frac{dy}{dt} = -k(y - y_s)$$

$$\Rightarrow \frac{1}{y-y_s}\,dy = -k\,dt$$

$$\Rightarrow \int \frac{1}{y-y_s}\,dy = \int -k\,dt$$

$$\Rightarrow \ln(y-y_s) = -kt + C$$

$$\Rightarrow y - y_s = e^{-kt+C}$$

$$\Rightarrow y - y_s = Ce^{-kt}$$

$$\Rightarrow y = y_s + Ce^{-kt}$$

$$y(0) = y_s + Ce^{-k\cdot 0}$$

$$\Rightarrow y_0 = y_s + Ce^{0}$$

$$\Rightarrow y_0 - y_s = C$$

$$\Rightarrow \boxed{y(t) = y_s + (y_0 - y_s)e^{-kt}}$$

Equilibrium:

$$\frac{dy}{dt} = 0$$

$$\Rightarrow -k(y - y_s) = 0$$

$$\boxed{y = y_s}$$

If y starts at y_s, it stays at y_s.

3.] Let $y(t)$ be the size of a population of interest at time t. Given limited resources or space to expand, many populations experience *logistic* growth at a per capita rate $k > 0$ to a *carrying capacity*, m:

$$\frac{dy}{dt} = ky\left(1 - \frac{y}{m}\right), \qquad y(0) = y_0.$$

Here, k and m are both strictly positive. Solve this ODE for $y(t)$ and determine the equilibria.

$$\frac{dy}{dt} = ky\left(1-\frac{y}{m}\right)$$

$$\Rightarrow \frac{1}{y\left(1-\frac{y}{m}\right)}\,dy = k\,dt$$

$$\Rightarrow \frac{1}{y} + \frac{1/m}{1-y/m}\,dy = k\,dt$$

$$\Rightarrow \int \frac{1}{y} + \frac{1/m}{1-y/m}\,dy = \int k\,dt$$

$$\Rightarrow \ln(y) - \ln(1-y/m) = kt + C$$

$$\Rightarrow \ln\left(\frac{y}{1-y/m}\right) = kt + C$$

$$\Rightarrow \frac{y}{1-y/m} = Ce^{kt}$$

$$\Rightarrow \frac{1-y/m}{y} = Ce^{-kt}$$

$$\Rightarrow \frac{1}{y} - \frac{1}{m} = Ce^{-kt}$$

$$\Rightarrow \frac{1}{y} = \frac{1}{m} + Ce^{-kt}$$

$$\Rightarrow y = \frac{1}{\frac{1}{m} + Ce^{-kt}}$$

$$\Rightarrow y = \frac{m}{1 + Cme^{-kt}}$$

$$\Rightarrow y(0) = \frac{m}{1 + Cme^{-k\cdot 0}}$$

$$\Rightarrow y_0 = \frac{m}{1 + Cme^{0}}$$

$$\Rightarrow \frac{m}{y_0} = 1 + Cm$$

$$\Rightarrow \frac{m}{y_0} - 1 = Cm$$

$$\Rightarrow \frac{1}{y_0} - \frac{1}{m} = C$$

$$\Rightarrow \frac{m-y_0}{y_0 m} = C$$

$$\Rightarrow y(t) = \frac{m}{1 + \frac{m-y_0}{y_0 m} m e^{-kt}}$$

$$\Rightarrow y(t) = \frac{m}{1 + \frac{m-y_0}{y_0} e^{-kt}}$$

$$\Rightarrow \boxed{y(t) = \frac{m y_0}{y_0 + (m-y_0)e^{-kt}}}$$

Equilibria:

$$\frac{dy}{dt} = 0$$

$$\Rightarrow ky(1-y/m) = 0$$

$$\boxed{y = 0 \text{ or } y = m}$$